

Shubham Kumar Sharma

Ajmer, RJ | shubhamsharma86900@gmail.com | +91-8690097812 | Portfolio | LinkedIn | Github

Summary

B.Tech IT student and Full-Stack Developer with strong foundations in **AI/ML**, **Cloud Computing**, and **Software Engineering**. Proficient in **Java**, **TypeScript**, and **Python**, with hands-on experience building scalable web and mobile applications using **React** and **React Native**. Backed by certifications from **AWS**, **IBM**, and **Cisco**, combining deep technical expertise in **Generative AI** and **AWS SageMaker** with modern frontend development to deliver robust, production-ready solutions.

Education

Engineering College, Ajmer, B.Tech in Information Technology | *CGPA: 9.06/10.0* Sept 2023 – Sept 2027
Coursework: Computer Architecture, Data Structure and Algorithms, Computational Theory, Computer Networking, O.S., Compiler Design, Software Engineering

AWS AI & ML Scholars – Future AWS AI Engineer Nanodegree Aug 2025 - Nov 2025
Sponsored by AWS in collaboration with Udacity
Coursework in Machine Learning, Generative AI, AWS SageMaker, and Bedrock, Terraform, Aurora Serverless PostgreSQL, S3

Government Moinia Islamia Sr. Sec. School, Senior Secondary | *Percentage: 91%* 2021 – 2023

Technical Skills

Languages: Python, C, Java, React Native, SQL	Frameworks: Spring, SpringBoot, React Expo
AI/ML Tools: AWS SageMaker, Bedrock, Jupyter Notebook, TensorFlow, PyTorch	Data/Cloud: APIs, Data Scraping, Data Cleaning, MySQL, PostgreSQL, IBM Cloud, AWS
Web/Other: HTML, CSS, JavaScript, Git, GitHub, Microservices	Tools: IntelliJ IDEA, VS Code, Git, GitHub, Docker, Postman, Maven

Experience

IBM SkillsBuild AI and Cloud Technologies Intern, Edunet Foundation – Remote July 2025 – Aug 2025

- Engineered and deployed a **multiclass classification model** achieving **95.8% accuracy** using **Batched Tree Ensemble (Snap Random Forest)** in IBM Watsonx.ai Studio.
- Applied **feature engineering**, **class balancing**, and **hyperparameter tuning** to optimize model performance.
- Automated the **data processing pipeline** (Pandas, Scikit-learn) for seamless ingestion, cleaning, training and evaluation.
- Designed a **prototype decision support tool** to accelerate applicant classification by **70%** for government benefit distribution.

Projects

Data Structures & Algorithms Visualizer Github

- Tools Used:** React, TypeScript, Vite, Tailwind CSS, Zustand, HTML5 Canvas, SVG.
- Engineered a custom **Animation Engine** that decouples algorithmic logic from UI rendering, ensuring a modular architecture and enabling stable 60 FPS animations.
- Optimized state management using **Zustand**, reducing component re-renders and improving application performance by 30% during complex algorithm executions.
- Implemented dynamic visualizations for **Arrays** and **Linked Lists** using **SVG** and **Canvas API**, facilitating real-time analysis of Time & Space Complexity.
- Designed a responsive, modern UI with **Tailwind CSS**, featuring granular playback controls (play, pause, step-through) that enhanced user interaction and learning efficiency.

INTELLIGENT DOCUMENT QUERYING SYSTEM (AWS BEDROCK) Github

- Tools Used:** AWS Bedrock, Terraform, Aurora Serverless PostgreSQL, S3, Python, Streamlit
- Built an end-to-end retrieval-augmented document querying system using AWS Bedrock Knowledge Bases and Aurora Serverless, enabling high-accuracy natural language search over PDF specifications.
- Designed and deployed cloud infrastructure using Terraform (VPC, subnets, NAT, Aurora cluster, S3, KB), delivering a scalable, secure, and production-ready GenAI application.

Quantum Circuit Simulator

- **Tools Used:** Python, Qiskit, Matplotlib, NumPy, Tkinter.
- Developed an interactive **Quantum Circuit Simulator** using **Python** and **Qiskit**, enabling accurate simulation and analysis of quantum algorithms and gate operations.
- Integrated **Matplotlib** to generate advanced visualizations such as **Bloch Spheres**, **State Vectors**, and probability histograms, simplifying abstract quantum mechanical concepts.
- Engineered a custom execution (transpiler) engine to support step-by-step circuit processing.
- Designed a modular interactive **GUI** with drag-and-drop functionality for quantum gates, significantly lowering the entry barrier for students learning Quantum Computing.
- Optimized simulation performance using **NumPy vectorization**, ensuring efficient processing of multi-qubit systems and complex circuit depths.

Certifications

- Future AWS AI Engineer Nanodegree, **Udacity & Amazon Web Services**, 2025
- AWS Educate Machine Learning Foundations – AWS, 2025
- Python Essentials 1 & 2 – Cisco, 2025
- Introducing Generative AI with AWS – Udacity, 2025
- Getting Started with Artificial Intelligence – IBM, 2025

Achievements

- Selected among top 1,500 scholars globally (from 50,000+ applicants) for the **AWS AI & ML Scholars – Future AWS AI Engineer Nanodegree (Scholarship)**, sponsored by Amazon Web Services (AWS) in collaboration with **Udacity**.
- Top 5% selection in **IBM SkillsBuild** AI & Cloud Technologies Internship.
- Completed 150+ coding problems on multiple coding platforms.